1. Introduction

The purpose of this Requirements Analysis and Design Document (RADD) is to detail the specifications for implementing a new **Business Process Automation (BPA) Platform**. This platform is designed to drive business value by automating key internal workflows and addressing critical inefficiencies identified in manual processes.

The current system suffers from long approval cycles, data entry errors, poor visibility into process status, and compliance risks due to missing audit trails, resulting in employee frustration and low productivity. This document serves as the blueprint for the automated workflow system, outlining the requirements, architecture, and design choices.

2. Requirements Specification

The requirements specification details the essential functions and expectations necessary for the successful implementation of the BPA Platform, covering business, stakeholder, functional, non-functional, and transition needs.

2.1 Business Requirements (BR)

These high-level requirements define what the system must achieve to deliver business value:

ID	Description	Source Problem Addressed
BR1	Minimize Cycle Time: The system must significantly reduce the end-to-end approval cycle time for core processes (e.g., procurement, expense claims).	Long approval cycles for procurement and expenses.
BR2	Improve Data Quality: The system must eliminate errors caused by manual data entry and validation.	Errors from manual data entry.
BR3	Ensure Compliance & Auditability: The system must provide a secure, digital, and immutable audit trail for all transactions and approvals.	Compliance risks due to missing audit trails.
BR4	Enhance Visibility: The system must offer centralized dashboards for tracking key performance indicators (KPIs) and identifying process bottlenecks.	Poor visibility into process status.
BR5	Boost User Productivity: The platform must provide an intuitive interface and mobile capability to reduce employee frustration and improve productivity.	Employee frustration and low productivity.

2.2 Stakeholder Requirements (SR)

The key participants in the work system are mapped to specific needs:

Stakeholder Group	Role in the BPA System	Key Stakeholder Requirements (SR)
Operations Staff	Process initiator (leave, expense claims) and real-time user.	SR1-OS1: As Operations Staff, I must be able to submit a request and capture/validate data online. SR1-OS2: As Operations Staff, I need real-time updates on my request status to plan my work.
Finance Staff	Approver (invoices/expenses) and recipient of real-time processing updates.	SR2-FS1: As Finance Staff, I need real-time updates and necessary financial data from the ERP integration to process requests. SR2-FS2: As Finance Staff, I need to easily access and generate audit logs and compliance reports.
Department Managers	Process Approver (leave, procurement) and process tracker.	SR3-DM1: As a Manager, I must receive automated notifications for pending tasks (email/SMS). SR3-DM2: As a Manager, I must be able to review and approve/reject requests via a secure online portal or mobile app.
Business Analysts	System administrator and performance monitor.	SR4-BA1: As a Business Analyst, I need access to the analytics dashboard to track bottlenecks and process performance. SR4-BA2: As a Business Analyst, I need the capability to define and modify process workflows and rules.
IT Administrators	System maintainer and technical support.	SR5-IT1: As an IT Admin, I need the system to be stable, scalable, and easy to maintain/update. SR5-IT2: As an IT Admin, I need access to system performance reports and audit logs for troubleshooting.

2.3 Functional Requirements (FR)

These requirements specify the actions the system must perform:

ID	Description
FR1	Request Submission: The system must allow employees to submit requests (leave, procurement, expense claim) through a centralized, web-based portal.
FR2	Data Capture & Validation: The system must capture all necessary data and validate it against predefined rules before routing.
FR3	Automated Routing Engine: The system must automatically route the request to the correct approver(s) based on predefined business rules.
FR4	ERP Integration: The system must integrate with the existing ERP (e.g., SAP/Oracle/Dynamics) to verify data and update finance/operations records in real-time.
FR5	Notification & Escalation: The system must send automated email/SMS notifications to approvers for pending tasks and escalate unaddressed tasks based on time limits.
FR6	Approval Action: The system must allow approvers to approve or reject a request via the online portal or a mobile application.
FR7	Process Dashboard: The system must provide a dashboard displaying the status of all processes, KPIs, and bottlenecks.
FR8	Audit Logging: The system must log every action, decision, and time stamp for audit purposes.

2.4 Non-Functional Requirements (NFR)

These requirements define system qualities such as performance, security, and reliability:

ID	Category	Description
NFR1	Performance	The system must be able to process a request submission and route it to the next approver within 2 seconds.
NFR2	Scalability	The system must be designed to handle high volumes of simultaneous requests (e.g., 500 concurrent users) without performance degradation.
NFR3	Security	The system must implement strong encryption protocols for data at rest and in transit to protect sensitive employee and financial information.
NFR4	Reliability	The system must have redundancy and failover mechanisms to ensure an operational uptime of 99.9%

		(NFR13 in guide).
NFR5	Usability	The mobile and web interface must be intuitive, requiring minimal user training (NFR7 in guide).
NFR6	Integrity	The system must maintain a detailed, immutable audit trail of all transactions and changes for compliance.

2.5 Transition Requirements (TR)

These requirements focus on the needs for a successful deployment (TR in guide):

ID	Description	Priority (as per guide logic)
TR1	Data Migration: Migrate all existing employee data, current purchase orders, and vendor master data into the new BPA Platform database.	High
TR2	System Integration Testing: Thoroughly test the integration points between the BPA platform and the ERP system (FR4).	High
TR3	Training: Provide comprehensive user training and support documentation for Operations, Finance, and Management staff.	Medium
TR4	Rollout: Map out a timeline and communicate the process of transmission and switchover from the legacy system to the new platform (TR14 in guide).	High

3. Requirements Verification & Validation

The verification and validation process ensures that the defined requirements are complete, consistent, and aligned with stakeholder expectations and business goals.

3.1 Requirements Verification (Check for Quality)

Requirements are verified against quality attributes like Unambiguous, Complete, and Scalability (as per Section 3.5 in guide).

REQ	Unambiguous	Complete	Functionality	Scalability	Integration
BR1	✓	✓	✓	✓	✓
FR4	✓	✓	✓	✓	✓
NFR2	✓	✓	✓	✓	✓
SR3-DM2	✓	✓	✓	✓	✓

- **Rationale:** All core requirements (Business, Functional, Non-Functional, Stakeholder) must pass the verification checks. For instance, **FR4 (ERP Integration)** is a key dependency mentioned in the source document and must be confirmed for *Integration*. **NFR2 (Scalability)** is critical to ensuring the new system doesn't create new bottlenecks and aligns with the need to handle high volumes, making the Scalability check essential.

3.2 Requirements Validation (Check for Alignment)

Requirements are validated against attributes like Feasibility, Correctness, Usability, and Reliability (as per Section 4 in guide).

REQ	Feasibility	Correct	Usability	Reliability	REQ
BR4	✓	✓	✓	✓	BR4
FR6	✓	✓	✓	✓	FR6
NFR4	✓	✓	✓	✓	NFR4
SR3-DM2	✓	✓	✓	✓	SR3-DM2

- **Rationale:** The validation confirms that the requirements are buildable and meet user needs. **SR3-DM2 (Mobile Approval)** is feasible given the availability of mobile device technology (Work System Document, Technologies section) and is essential for Usability for managers on-the-go. **NFR4 (99.9% Uptime)** directly addresses the need for a reliable platform to prevent a return to manual processes, requiring a **Reliability** check.

4. Architecture and Design requirements

4.1 Use Case Diagram (Conceptual)

The primary system actors and their core interactions:

Actor	Use Case	Description
Employee (Operations Staff/Finance Staff)	Submit Request	The user creates and submits a new Leave, Procurement, or Expense Claim.
Department Manager	Approve/Reject Request	The manager receives a notification and takes an action (Approve/Reject) on a pending request, potentially via a mobile device.
System (BPA Platform)	Route Request	The platform automatically validates data and forwards the request to the next appropriate approver based on predefined rules.
Business Analyst/Admin	Monitor Dashboard	The user accesses the centralized reporting dashboard to track process performance and KPIs.

System (BPA Platform)	Log Audit Trail	The system securely records all transaction steps for compliance.

4.2 System Design Components (Section 6.2 in Guide)

The system architecture will be composed of the following core components, based on the technology requirements specified in the source document:

1. **Presentation Layer (Client Interface):**
    - **Web Portal:** Interface for request submission and dashboard access (used by all internal customers).
    - **Mobile App:** Interface for on-the-go approvals and notifications (used primarily by Management).
2. **Application/Logic Layer (BPM Engine):**
    - **Workflow Automation Software:** This is the core engine (e.g., UiPath, Power Automate, or other BPM software) responsible for rule execution, routing, notifications, and escalations (FR3, FR5).
    - **Data Validation Module:** Component that performs checks on submitted data (FR2).
3. **Data Layer:**
    - **Database Management System:** Stores all process workflows, audit logs, employee details, and process data.
    - **Audit Log Store:** Dedicated secure storage for compliance records (FR8).
4. **Integration Layer (Connectors):**
    - **ERP Connector:** Secure, two-way API connection to the existing ERP system (SAP/Oracle/Dynamics) for data synchronization and real-time updates to Finance/Operations.
    - **Reporting Connector:** Interfaces with Reporting & Analytics tools (e.g., Power BI) to feed dashboard data (FR7).

5. Design Options (Section 6 in Guide)

The implementation approach must be assessed against common design options for an enterprise-level automation platform: Purchasing (Outsourcing), Licensing, or Internal Development. Given the

requirement for deep integration with the existing ERP system and the need to define custom process rules, a purely outsourced or licensed off-the-shelf solution may be complex.

Design Option	Pros	Cons	Recommended Choice
Purchasing (Outsourcing)	Fastest implementation time; Reduced internal maintenance burden.	High vendor lock-in; Customization for ERP integration (FR4) can be costly.	No
Licensing (COTS Platform)	Access to proven, mature BPM software technology (e.g., UiPath, Power Automate); Lower initial development cost than internal build.	Ongoing license fees; Limited ownership of Intellectual Property (IP); May require significant configuration for custom business rules.	Yes
Internal System Development	Full control over customization and IP ownership; Perfect alignment with unique business requirements (BR1-BR5).	Highest initial development cost; Longest time to market; Requires specialized internal IT resources (SR5-IT1).	No

Choice: The **Licensing** of a Commercial Off-The-Shelf (COTS) Business Process Management (BPM) platform (like UiPath or Power Automate) is the recommended approach. This provides a balance, ensuring the use of industry-standard workflow tools while allowing for the necessary customization and integration required for FR4 and the specific process rules outlined in the source document. The platform will serve as the engine, with internal IT teams building the connectors and custom rules required.

6. Financial Analysis (Conceptual)

This section forecasts the costs and potential value of the selected licensing option over a four-year period, based on a typical enterprise software adoption model.

6.1 Potential Value of Licensing (Section 7.2 in Guide)

The primary benefit is calculated based on the reduction of employee time spent on manual process tasks and the reduction of errors (BR1, BR2).

Year	Costs (License & Support)	Benefits (Productivity/Error Reduction)	Net Value
Year 1	High (Implementation + Annual License)	Low (Initial rollout, 20% process coverage)	Negative
Year 2	Medium (Annual License + Minor Customization)	Medium (50% process coverage, staff trained)	Positive
Year 3	Medium (Annual	High (80% process	High Positive

	License)	coverage, optimized performance)	
Year 4	Low (Annual License + Minimal Support)	High (95% process coverage, sustained gains)	High Positive

Explanation of Costs

- **Initial Costs:** Include procurement of the BPM software license, professional services for initial setup and ERP integration (FR4), and one-time hardware upgrades for hosting servers (if required).
- **Recurring Costs:** Primarily the annual software licensing fees, cloud storage fees for audit logs, and internal IT administrator salaries for system maintenance and support (SR5-IT1).

Explanation of Benefits

- **Efficiency Gains (BR1):** Reduced employee time spent on manual routing and follow-ups due to automated workflows.
- **Error Reduction (BR2):** Financial savings derived from eliminating costly manual data entry errors and rework.
- **Compliance Risk Mitigation (BR3):** Reduced risk of fines or sanctions due to missing audit trails, a critical risk identified in the problem statement.
- **Strategic Value (BR4, BR5):** Improved managerial decision-making due to real-time process visibility and increased employee job satisfaction and productivity.